AYE
TECH HUB

• MECHANICAL ENGINEERING • LESSON 03

Engineering Mechanics

Statics · FBD · Equilibrium · Moments · Dynamics

Lesson 03 of 30 | Mechanical Engineering Program

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- Newton's 3 laws — the foundation of every mechanics calculation
- Force vector resolution: $F_x = F \cos \theta$, $F_y = F \sin \theta$
- Free body diagrams and the 3 equilibrium equations ($\Sigma F_x = \Sigma F_y = \Sigma M = 0$)
- Pin, roller and fixed supports — reactions and determinacy
- Moments, friction and introduction to dynamics ($F=ma$, energy)

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Forces, Vectors and Newton's Laws

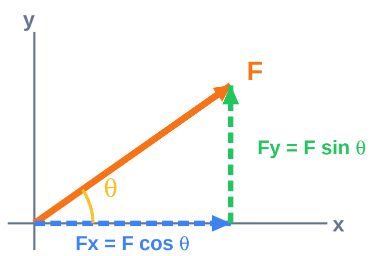
Engineering mechanics is the science of **how forces cause motion — or prevent it**. It divides into two branches: **Statics** (bodies in equilibrium — no acceleration) and **Dynamics** (bodies in motion under unbalanced forces). Every structural calculation, every machine design, and every vehicle analysis begins here.

Newton's Three Laws — The Foundation

Law	Statement	Engineering Application
1st Law (Inertia)	A body remains at rest or in uniform motion unless acted on by a net force.	Static equilibrium: $\Sigma F = 0$ and $\Sigma M = 0$. A beam stays put because reactions balance loads.
2nd Law ($F = ma$)	Net force = mass × acceleration. The direction of acceleration equals the direction of net force.	Dynamic analysis: sizing motors, calculating stopping distances, vehicle dynamics.
3rd Law (Action-Reaction)	Every action force has an equal and opposite reaction force.	Support reactions: a floor exerts an upward force equal to the weight placed on it.

Force Vector Resolution

A force is a **vector quantity** — it has both magnitude and direction. To analyse structures, we always resolve forces into their horizontal (x) and vertical (y) components. This converts a force at any angle into two perpendicular forces that are easy to sum algebraically.



$F = \sqrt{F_x^2 + F_y^2}$

$\theta = \arctan(F_y/F_x)$

$F_x = F \cos\theta$

$F_y = F \sin\theta$

Figure 3 — Force Vector Resolution into X and Y Components

Quantity	Symbol	Unit	Vector or Scalar?	Examples
Force	F	Newton (N)	Vector	Weight, tension, reaction, friction
Moment/Torque	M	Newton-metre (N·m)	Vector	Turning a bolt, beam bending moment
Mass	m	Kilogram (kg)	Scalar	Mass of a component
Weight	W	Newton (N)	Vector	$W = mg$, acting downward at CoG
Pressure	p	Pascal (Pa = N/m²)	Scalar	Hydraulic pressure, wind pressure
Stress	σ	Pa or N/mm² (MPa)	Tensor	Internal force per unit area in material
Acceleration	a	m/s²	Vector	$g = 9.81$ m/s² (gravitational)

Sign convention: Always define positive directions at the start of every problem. Common choice: right = +x, up = +y, anti-clockwise = +moment. Stick to this throughout — a sign error is the most common source of wrong answers in statics.

Equilibrium and Free Body Diagrams

A structure is in **static equilibrium** when the sum of all forces and the sum of all moments about any point are both zero. The three equilibrium equations are the most-used tools in structural analysis: **$\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma M = 0$**.

A **Free Body Diagram (FBD)** is an isolated sketch of the structure showing every external force and moment acting on it. Drawing a correct FBD is the single most important step in solving any statics problem — it makes the physics visible before you write a single equation.

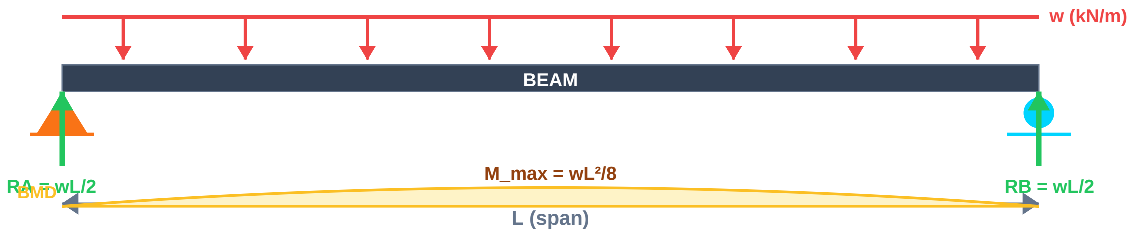


Figure 1 — Free Body Diagram: Simply Supported Beam with UDL

How to Solve a Beam Problem — Step by Step

Step	Action	Example (UDL beam, span L, load w kN/m)
1	Draw the FBD — isolate the beam, show all loads and unknown reactions	Show UDL arrows above beam; RA↑ at left pin; RB↑ at right roller
2	Apply $\Sigma F_y = 0$ (vertical equilibrium)	$RA + RB - w \times L = 0 \rightarrow RA + RB = wL$
3	Apply $\Sigma M = 0$ about left support (eliminates RA)	$RB \times L - (w \times L) \times (L/2) = 0 \rightarrow RB = wL/2$
4	Substitute back to find RA	$RA = wL - wL/2 = wL/2$ (symmetric as expected)
5	Apply $\Sigma F_x = 0$ to find horizontal reactions	$RA_x = 0$ (pin horizontal reaction = 0 if no horizontal loads)
6	Draw shear force and bending moment diagrams (SFD/BMD)	$V(x) = RA - wx$; $M(x) = RA \times x - w \times x^2/2$; $M_{max} = wL^2/8$ at midspan

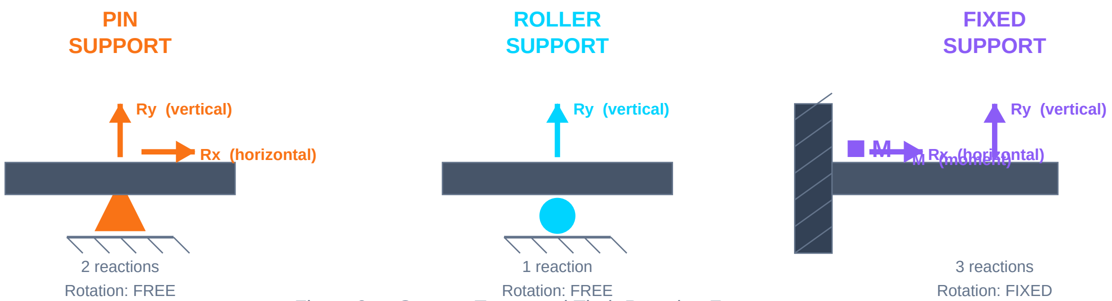


Figure 2 — Support Types and Their Reaction Forces

Degree of indeterminacy: A structure is statically determinate if the number of unknowns equals the number of equilibrium equations. A pin adds 2 unknowns; a roller adds 1; a fixed support adds 3. A simply supported beam (pin + roller) has 3 unknowns and 3 equations — perfectly determinate.

Moments, Friction and Introduction to Dynamics

A **moment** (or torque) is the turning effect of a force about a point. Moment = Force × perpendicular distance to the pivot. This principle governs everything from a spanner tightening a bolt to a crane lifting a load.

Moments and Couples

Concept	Formula	Notes and Application
Moment of a force	$M = F \times d$ (d = perpendicular distance)	Units: N·m. Sign: clockwise = negative (common convention). Larger lever arm = same force creates bigger moment.
Couple	$M = F \times d$ (two equal opposite forces)	A couple creates pure rotation with no net force. Steering wheel, propeller torque, tightening a tap.
Varignon's Theorem	$M_{total} = \Sigma F_i \times d_i$	Moment of a resultant = sum of moments of component forces. Simplifies complex distributed load problems.
Bending Moment (beam)	$M(x) = \Sigma M$ of all forces left of section x	Max bending moment → max bending stress. $\sigma_{max} = M_{max} \times y_{max} / I$ (flexure formula).
Torque (shaft)	$T = P / \omega = 9\,550 \times P(kW) / n(RPM)$	Power and speed determine shaft torque. Used to size shaft diameter and key/spline connections.

Friction

Type	Formula	Coefficient (μ)	Key Rule
Static friction (resists motion start)	$F_{s\_max} = \mu_s \times N$	μ_s: Steel on steel 0.15–0.3 Rubber on concrete 0.6–0.8	Object does not move until applied force exceeds F_{s_max}
Kinetic / sliding friction (opposes sliding motion)	$F_k = \mu_k \times N$	$\mu_k < \mu_s$ always (kinetic is always less than static)	Once sliding starts, friction force is constant regardless of speed
Rolling resistance	$F_r = C_r \times W$	C_r: Steel on steel 0.001–0.002 Tyre on road 0.01–0.03	Much smaller than sliding friction; why wheels and bearings save energy

Introduction to Dynamics

Topic	Key Equations	Application
Kinematics — linear (uniform acceleration)	$v = u + at$ $s = ut + ½at^2$ $v^2 = u^2 + 2as$	Stopping distance of a vehicle; conveyor belt acceleration; projectile trajectory
Newton's 2nd Law	$F_{net} = m \times a$ $F_{net} = ma$ (vector)	Sizing motor to accelerate a load; force needed to stop a moving mass
Work and Energy	$W = F \times d$ (J) $KE = ½mv^2$ $PE = mgh$ $W_{net} = \Delta KE$	Energy balance in mechanisms; impact loading; flywheel energy storage
Power	$P = W/t = F \times v$ (W = Watts)	Motor sizing; pump power; conveyor drive requirements
Impulse-Momentum	F × t = m × (v■ – v■)	Crash analysis; hydraulic hammer impact; braking force calculation
Rotational dynamics	$T = I \times \alpha$ $I = \Sigma mr^2$ (moment of inertia)	Flywheel design; gear ratio selection; rotating shaft dynamics

What comes next — MECH-004: Thermodynamics Basics — the laws of thermodynamics, heat transfer modes, the Carnot cycle, and how thermal energy analysis underpins engines, refrigeration, and power systems.